The EAR Radio Frequency Interference Detection and Geolocation Methodology: Application to the NASA-ISRO Synthetic Aperture Radar (NISAR) Geocoded Polarimetric Covariance Product

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***Abstract*—The National Aeronautics and Space Administration (NASA) Indian Space Research Organization (ISRO) synthetic aperture radar (SAR) (NISAR) mission February 2026 public data release includes 13 803 Level-2 Geocoded Polarimetric Covariance (GCOV) granules reporting measured normalized radar cross sections (NRCSs). An investigation of L-band radio frequency interference (RFI) in the GCOV product is presented that demonstrates the susceptibility of the mission's observations to anthropogenic transmissions. An "Elliptical ARtifact" (EAR) RFI detection and geolocation methodology is described and applied to NISAR GCOV products. Because GCOVs represent multilooked measurements, existing RFI detection methods that process "raw" in-phase and quadrature backscatter returns are not applicable. The EAR method instead performs detections based on the "shape" of backscatter intensity anomalies consistent with the properties of NISAR's SweepSAR mode of operation. Results in the vicinity of known sources of interference are used to demonstrate the performance of the EAR method. Larger scale tests further reveal 2396 detections representing more than 500 distinct sources across locations included in the February 2026 public data release. The majority of the sources identified in this "proof of concept" demonstration do not appear to have been previously reported, so the need for RFI detection beyond that available from a priori information on potential RFI sources is emphasized. The use of NISAR's multiple operational modes in characterizing RFI source properties is also demonstrated. While future NISAR data releases are expected to include additional mission-provided RFI detection and mitigation, the EAR technique will remain useful for independent assessment of mitigation performance and for identifying residual RFI corruption in the NRCS products of NISAR or other SAR missions.**

***Index Terms*—L-Band, microwave remote sensing, National Aeronautics and Space Administration (NASA)-Indian Space Research Organization (ISRO) synthetic aperture radar (SAR) (NISAR), radio frequency interference (RFI), SAR.**

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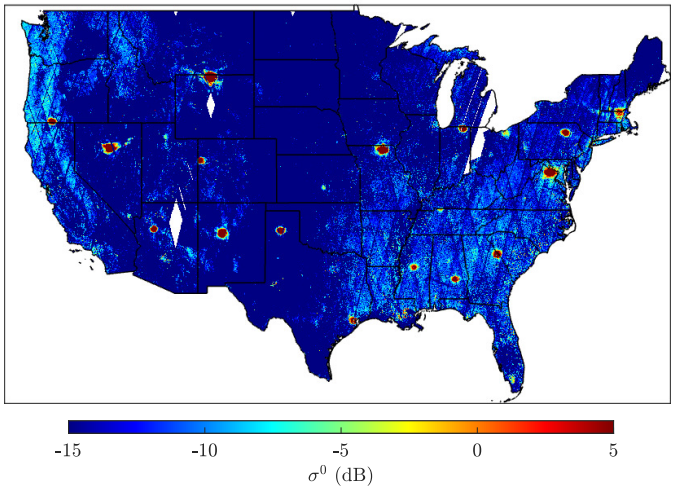


Fig. 1. Mean HH polarized NRCS derived from NISAR's "frequency B" observations using acquisitions made over the period December 2025–January 2026 across the CONtiguous United States included as part of the February 2026 public data release.

I. Introduction

THE National Aeronautics and Space Administration (NASA)-Indian Space Research Organization (ISRO) synthetic aperture radar (SAR) [1] (NISAR), launched July 30, 2025, is advancing Earth monitoring capabilities at L- and S-bands by providing measurements with more frequent revisit at finely resolved spatial resolutions of Earth's backscatter. It has long been recognized that radio frequency interference (RFI) can impact the operation of spaceborne microwave sensors. Within the L-band portion of the radio spectrum (1–2 GHz), extensive RFI investigations have been reported for microwave radiometers operating 1400–1427 MHz [2], [3], [4], [5], [6], [7], [8], a portion of the spectrum reserved by the Federal Communications Commission (FCC) and International Telecommunication Union (ITU) for the "Earth exploration satellite service-passive" service in which transmissions by other systems are prohibited. Similar RFI investigations have been conducted for GNSS-R systems [10], [11], [12] operating in the 1559–1610-MHz range

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reserved for "radionavigation satellites" transmitting from space toward Earth's surface. Unlike these systems, NISAR uses frequencies in the 1215–1300-MHz range, with the exact frequencies occupied varying dynamically depending on the radar mode at the time of observation. While 1215–1240 MHz is allocated primarily for "Earth exploration satellite service-active" sensing systems such as NISAR, the remainder is shared with other applications including "radiolocation" (i.e., terrestrial surface-to-air radars). Given these allocations, NISAR must contend with numerous other civilian and defense systems that have equal right to transmit in the bands used for its observations. Although the pulse compression and SAR processing associated with NISAR's data products provide benefits in reducing RFI impacts as compared to microwave radiometry and GNSS-R systems, the wide prevalence of RFI in NISAR measurements motivates careful studies of RFI corruption and reduction of its effects.

Past studies of SAR RFI detection and mitigation have emphasized techniques applied to the "raw" in-phase and quadrature components of the received signal [13], [14], [15], [16] before pulse compression, SAR processing, and any multilooking. A subset of related techniques may be applied to NISAR products that include "Level-0 Radar Raw Signal Science Data" or "Level-1 Range Doppler Single Look Complex." In contrast, the NISAR L-band Level-2 Geocoded Polarimetric Covariance (GCOV) [17] product of interest in this work reports measured normalized radar cross sections (NRCSs) that have already been SAR processed and multilooked into polarimetric covariance products. An initial indication of the susceptibility of NISAR's L-band measurements to the presence of interference is provided in Fig. 1, which depicts mean HH-polarized NRCS levels across the contiguous United States for the period December 2025 to January 2026 from the February 2026 GCOV data release. The 20-dB anomalies observed above background levels correspond to interfering signals originating from the known locations of FAA long-range air traffic control radar systems. While the 18 "hotspots" in Fig. 1 clearly indicate the significant impacts of RFI, these examples serve only as initial qualitative indicators of interference that is large enough to be obvious in Fig. 1. A significantly larger number of interference sources than those evident in Fig. 1 in fact impact NISAR's measurements, as will be shown in what follows. The prevalence of RFI motivates sensitive detection methodologies capable of identifying even low-level image corruption that remains significant enough to bias science results. The further geolocation of RFI sources is motivated to aid in developing spatial masks that flag and address RFI contamination not only when it is large but also in areas surrounding sources where corruption is of reduced amplitude.

It is also noted that as part of NISAR's ongoing calibration and validation phase, efforts relating to improving estimates of the elevation antenna pattern (EAP), enhancing geolocation accuracy as well as valid sample masking are all ongoing. Refinements to the mission's "raw-data" RFI detection and mitigation processor are also ongoing, with the updated processor expected to be applied in future data releases. The results shown here therefore represent RFI detection and geolocation without application of the mission's RFI mitigation strategy. The proposed procedures are expected to remain applicable for future NISAR data releases for detecting and assessing any residual RFI GCOV data corruption and for identifying the associated RFI sources.

Section II provides a brief overview of the NISAR's acquisitions used followed by a description of the proposed detection and geolocation method in Section III. Site-specific and global detection studies obtained by applying the proposed methods to the 13 803 GCOV acquisitions of the February 2026 release are described in Section IV. A summary of the observed trends, practical considerations, and other conclusions are then presented in Section V. The presented results consider only the February 2026 L-band data release; similar studies for the mission's S-band observations will be reported in future work.

II. Background

NISAR is a highly capable radar that operates under an observation plan that varies measurement properties for each track/frame (i.e., location) and on ascending and descending passes; for a comprehensive description, see [18], [19], [20], [21], [22]. Acquisition properties are tailored to maximize scientific utility while balancing data volume and duty cycle constraints. Measurement properties that are varied include the transmitted chirp's duration (τ_c), bandwidth (B_w), center frequency (f_c), and polarization properties of the measured NRCS (σ^0_{pq}) both on transmit (p) and receive (q). Properties of the February 2026 GCOV public data release pertinent to this work are summarized in Fig. 2. The summary in Fig. 2(a) displays modes with more varied polarizations first; note that at a given location, multiple modes may occur on distinct ascending or descending passes. For the February 2026 release, 5.79%, 6.47%, 84.76%, 0.33%, and 2.67% of locations are viewed with the modes single-pol H (SH), single-pol V (SV), dual-pol HH/HV (DH), dual-pol VV/VH (DV), and quad-pol (QP), respectively. Of particular interest are the acquisition's center frequency and bandwidth as depicted in Fig. 2(b)–(f), which report the number of granules in the February 2026 release available in five combinations. To support ionospheric delay correction, over most scenes NISAR concurrently acquires measurements at two distinct center frequencies nominally designated as frequencies "A" and "B." Because the center frequency and bandwidth of frequencies "A" and "B" vary depending on the acquisition mode, the A/B designations are less relevant. What is of particular interest is the specific bandwidth and center frequency used since an interference signal will only impact NISAR's acquisitions if its spectral extent coincides with those of NISAR's at a given time and location.

Five measurement configurations are identified and referred to in what follows as $\mathcal{M}_1$-$\mathcal{M}_5$; the corresponding measurement bandwidth and center frequency combinations are detailed in Table I. An examination of Fig. 2(b)–(f) makes clear that measurement $\mathcal{M}$ varies by location where for example $\mathcal{M}_3$ performs measurements with B_w = 40 MHz and f_c = 1.239 GHz that are primarily available over North

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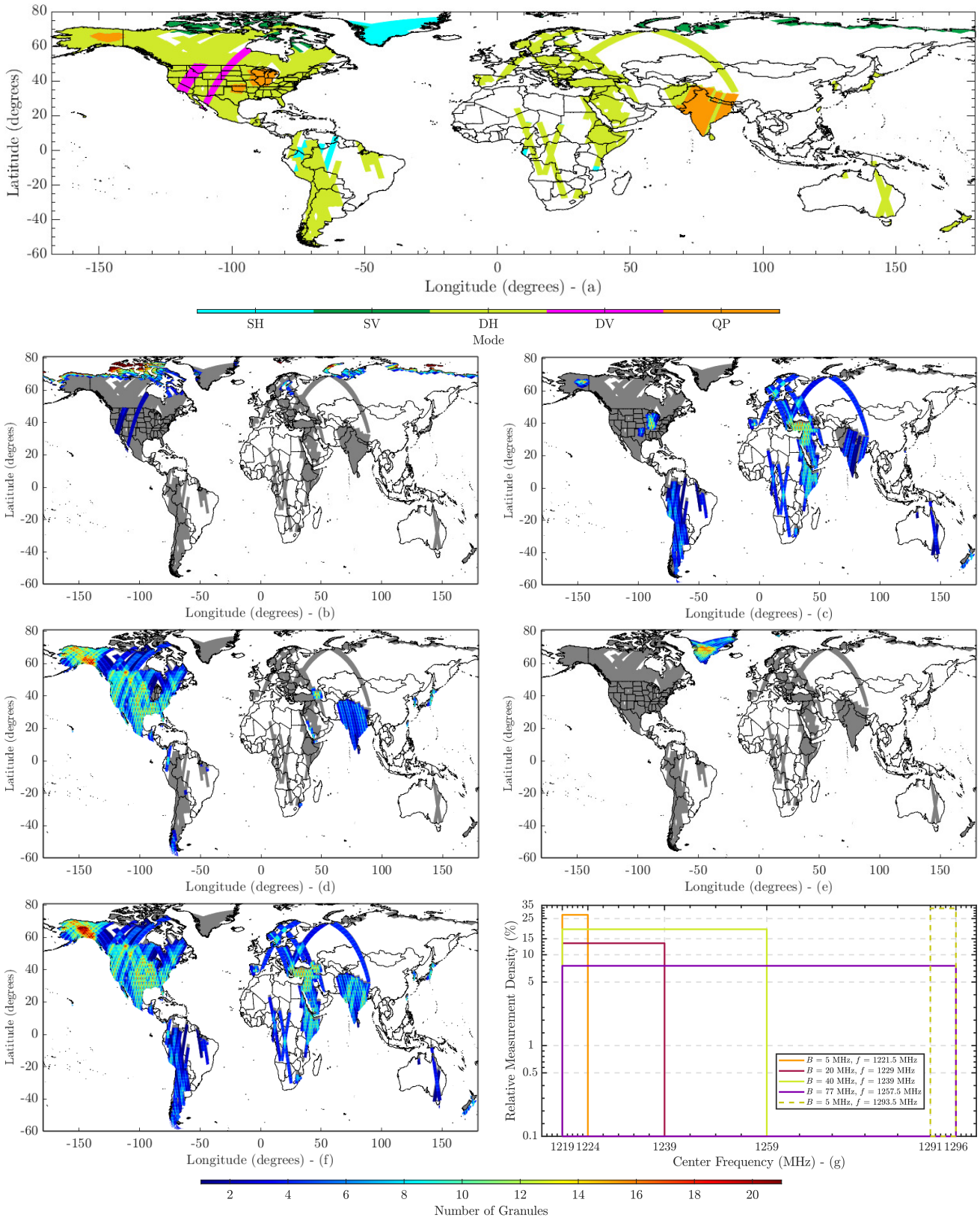


Fig. 2. Properties of February 2026 public GCOV data release. View is limited to land. (a) Ranked acquisition mode projected on 5-km grid. (b) Granule density, $B = 5$ and $f = 1221.5$ MHz. (c) Granule density, $B = 20$ and $f = 1229.0$ MHz. (d) Granule density, $B = 40$ and $f = 1239.0$ MHz. (e) Granule density, $B = 77$ and $f = 1257.5$ MHz. (f) Granule density, $B = 5$ and $f = 1239.5$ MHz. (g) Measurement configuration spectral contents.

America and India, while $\mathcal{M}_2$ with $B_w = 20$ MHz and $f_c = 1.229$ GHz is more widely available over South America and other regions of the world (excepting North America) in the February 2026 data release. Fig. 2(g) shows that while

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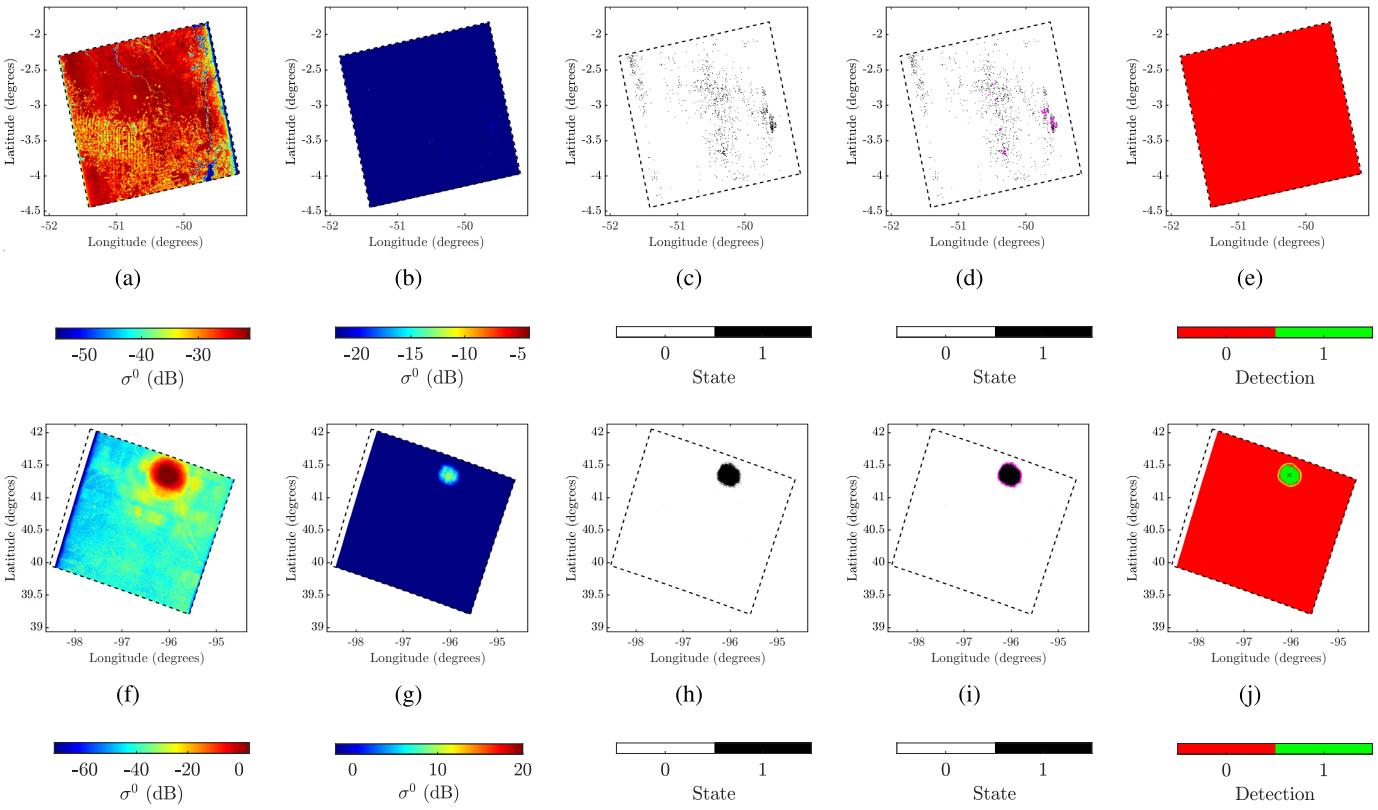


Fig. 3. Example intended to illustrate the intermediate steps associated with the application of the EAR RFI detection and geolocation methodology. The first row relates to an acquisition made over the Amazon River on December 21, 2025 where no interference source is detected. The second row relates to an acquisition made on December 2, 2025 over Omaha, Nebraska, USA, where an interference source is detected. Both are cross polarized σ^0_{hv} estimates. (a) σ^0_{hv} with colorscale limited to the 1st and 99th percentiles. (b) σ^0_{hv} with colorscale limited to the 98th and 100th percentiles. (c) Binarized mask. (d) Binarized mask with segmentation polygons in magenta overlaid. (e) Illustration of detection state along with potential source location after application of circularity test to segmented polygons. (f) σ^0_{hv} with colorscale limited to the 1st and 99th percentiles. (g) σ^0_{hv} with colorscale limited to the 98th and 100th percentiles. (h) Binarized mask. (i) Binarized mask with segmentation polygons in magenta overlaid. (j) Illustration of detection state along with potential source location after application of circularity test to segmented polygons.

$\mathcal{M}_2$ spans $1219 \leq f \leq 1239$ MHz, $\mathcal{M}_3$ spans $1219 \leq f \leq 1259$ MHz (i.e., $\mathcal{M}_2$ is a spectral subset of $\mathcal{M}_3$). As a result, $1219 \leq f \leq 1239$ MHz sources of interference located in India and South America would appear in both $\mathcal{M}_2$ and $\mathcal{M}_3$ measurements. A $1239 < f \leq 1259$ MHz source in contrast would impact NISAR's measurements only in India given the $\mathcal{M}_3$ measurements that are not present in South America. The requirement to interpret NISAR RFI properties in the context of the NISAR observation plan and its variation in space combined with source frequency occupancy is thus evident.

III. Proposed Methodology

NISAR employs a novel "SweepSAR" technique that enables a ≈240-km swath that is achieved, in part, through the use of digital beamforming along range using 12 dedicated L-band transmit/receive modules so that an approximate 20-km footprint extent in range occurs. The extent of the instantaneous footprint along Earth's surface then varies with range, given the projection onto the ground and the varying incidence angles. These effects are modest across NISAR's swath, so that an approximate 20-km along-ground extent is assumed in what follows. The extent of each footprint in cross range is determined by the ratio of the wavelength to the observatory's 12-m reflector antenna aperture. When combined with NISAR's 750 km orbital altitude, a cross-range footprint size of approximately 20 km also results. Regardless of the exact dimensions of each individual footprint, the footprint may be thought of as an ellipse with a high axial ratio. Following range compression and SAR processing, an in-band interference source will typically act to raise a measurement's noise floor throughout the entirety of the antenna footprint within which it lies. This then results in antenna footprint-shaped anomalies surrounding an RFI source's location. It follows that a majority of terrestrial interference signals will manifest in GCOV products as ellipses with high axial ratios (i.e., quasi-circular regions). The proposed Elliptical ARtifact (EAR) RFI detection and geolocation methodology implements a multistep procedure to identify features consistent with the presence of RFI.

The step-by-step procedure is summarized in Fig. 3 using example RFI-free [plots (a)–(e)] and contaminated [plots (f)–(j)] examples. The colorscale in plots Fig. 3(a) and (f) extends from the acquisitions' respective 1st and 99th percentile. The procedure limits its analysis to the subset of pixels whose amplitudes exceed the scene 98th percentiles (recall that interference acts to raise σ^0). While lower cutoffs may allow the detection of lower powered interference sources and higher cutoffs would have the effect of limiting detections

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TABLE I
UNIQUE MEASUREMENT BANDWIDTH/FREQUENCY COMBINATIONS AVAILABLE AS PART OF THE FEBRUARY 2026 PUBLIC DATA RELEASE

Configuration	B_w (MHz)	f_c (MHz)	Density (%)
$\mathcal{M}_1$	5	1221.5	27.50
$\mathcal{M}_2$	20	1229.0	13.38
$\mathcal{M}_3$	40	1239.0	19.09
$\mathcal{M}_4$	77	1257.5	7.55
$\mathcal{M}_5$	5	1293.5	32.48

to only the most egregious sources, for this initial demonstration the 98th percentile was found to be adequate. The definition of this threshold in percentile terms makes the detection procedure less dependent on a scene's natural amplitude variability as well as any potential calibration uncertainties. The impact this has on perceived relative σ^0 distributions is depicted in Fig. 3(b) and (g).

Subsequently, the acquisitions are binarized with pixels whose amplitudes are greater than the 98th percentile assigned the logical state of 1, while those below the threshold are assigned a state of zero. A segmentation algorithm to the binarized data is then applied to identify and isolate contiguous and unique features within the image [see polygons in Fig. 3(d) and (i)]. Because this is based on the relative 98th percentile cutoff, all scenes irrespective of the presence of an RFI source will retain features for segmentation. For computational efficiency, only the largest 20 segmented features are considered for further analysis. For each of the polygons depicted in plots Fig. 3(d) and (i), an ellipticity test is then applied. The properties of the individual footprints associated with NISAR's "SweepSAR" mode of operation together with trends observed across the 13 803 public granules motivate applying the circularity test

$$\odot = \frac{4\pi A}{p^2} \tag{1}$$

where $\odot$ is the measure of circularity, A is the area of the polygon, and p is its perimeter. Given the $A = \pi r^2$ of a circle together with its $p = 2\pi r$ perimeter, the higher the $\odot$ metric, the more circular the region segmented is expected to be with $\odot = 1$ corresponding to a perfect circle.

In the case of the RFI-free scene over a tributary of the Amazon River in plots Fig. 3(a)–(e), the mean $\odot$ is 0.25 and no detections are made. Conversely, the clear indications of an active interference source in Fig. 3(f)–(j) give rise to $\odot = 0.94$ for which detection is recommended. As a final step, an attempt to geolocate the more precise location of the source is made by finding centroids of any regions recommended for detection. As the subset of pixels within a "detection polygon" forms a cluster, its centroid position is estimated using

$$\phi_T = \frac{\sum_{i=1}^{N_C} \phi_g(i) \cdot \sigma^0_{\mathrm{pq}}(i)}{\sum_{i=1}^{N_C} \sigma^0_{\mathrm{pq}}(i)} \tag{2}$$

$$\theta_T = \frac{\sum_{i=1}^{N_C} \theta_g(i) \cdot \sigma^0_{\mathrm{pq}}(i)}{\sum_{i=1}^{N_C} \sigma^0_{\mathrm{pq}}(i)}. \tag{3}$$

where N_C is the number of pixels within a cluster and σ^0_{pq} is each pixel's NRCS, denoted by the latitudes ϕ_g and longitudes θ_g. The resulting estimate of the transmitter's position at latitude θ_g and longitude ϕ_g is then ≈41.34°N ≈96.04°W. The source of the interference for the acquisition in plots Fig. 3(f)–(j) is found to originate from the North Omaha Airport and the "retrieved" centroid is within 2 km of its pin-point location, where the operation of airport surveillance radars (ASRs) is to be expected.

While tuning and further testing of detection thresholds is expected as more data become available, initial tests across the 13 803 February 2026 public granules suggest that a combination of $\odot$ and A thresholds is useful for maximizing detection and minimizing false alarm rates. Because a subset of natural scattering signatures (e.g., mountain peaks) will also exhibit circular signatures, detections are limited to features whose area is sufficiently large. Initial testing suggests the multiple regions $\odot \geq 0.85$ and $\sqrt{A} \geq 3$ km, $0.75 \leq \odot < 0.85$ and $\sqrt{A} \geq 5$ km, and $0.5 \leq \odot < 0.75$ and $\sqrt{A} \geq 8$ km as representative of RFI. The final combination often coincides with higher powered transmitters whose signatures may overlap in multiple antenna footprints, thereby giving rise to a lower $\odot$ value that extends over a larger region.

IV. OBSERVED TRENDS

The EAR method applied to the February 2026 L-band release provides a rich dataset to investigate the properties of interference sources, as described in this section.

A. Transient Versus Persistent Sources

NISAR's 12-day revisit offers the opportunity to examine the repeatability of detections and the "persistence" of particular RFI sources over the period December 1, 2025 to January 18, 2026 of the February 2026 release. An example is depicted in Fig. 4 using three acquisitions on December 7, 19, and 31 over Turkey, all having $B_w = 20$ MHz and $f_c = 1.229$ GHz. The EAR algorithm detects RFI in all three images; Fig. 4(a)–(c) shows clear artifacts having $0.85 < \odot < 0.9$ together with amplitude anomalies more than 15 dB above the scene medians. Repeated detection of the "left" source at 37.92°N, 40.20°E is also evident. The source in this case is an ASR at Diyarbakır Airport. While detections are also made at 37.95°N, 41.15°E (the "right" source), they occur only for the December 7 and 31 acquisitions, and the source is not evident in Fig. 4(b). This source is found to exist at the location of Batman Airport, Turkey. While the underlying causes for the transience of the Batman Airport source are not known, such variability in a radar source may be associated with its antenna rotation in azimuth, which typically has a rotational period of 4–6 s. Because the cross-range extent of an NISAR antenna footprint may be spanned within a time less than the antenna rotation rate, measurements with the ASR antenna directed opposite NISAR will experience reduced RFI. An interruption in its operation may be another factor.

A second example is depicted in Fig. 5 for a series of acquisitions made on December 6, December 23, and January 11th.

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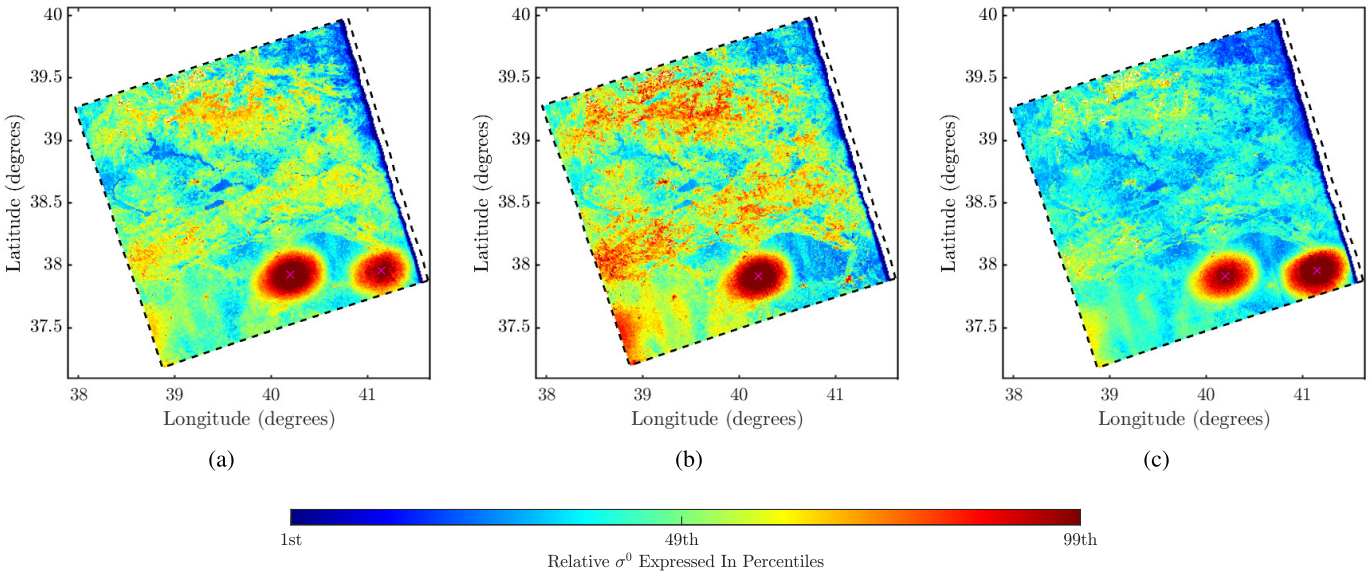


Fig. 4. Series of $\mathcal{M}_2$ acquisitions made over Turkey highlighting "transience" of one of the two sources. (a) December 7, 2025. (b) December 19, 2025. (c) December 31, 2025.

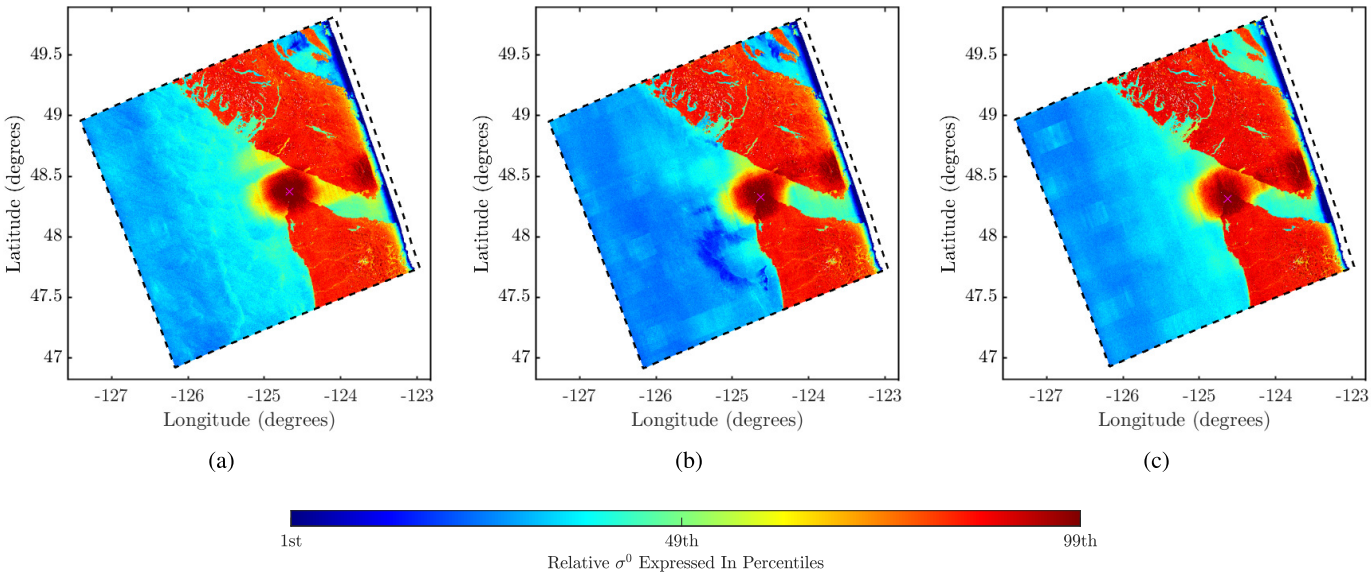


Fig. 5. Series of $\mathcal{M}_3$ acquisitions made over Classet, Washington, USA, highlighting "persistence" of interference source. (a) December 6, 2025. (b) December 23, 2025. (c) January 11, 2026.

All acquisitions have a B_w = 40 MHz and f_c = 1.239 GHz. In this case, the detections at 48.37°N 124.66°W in Classet, Washington, USA, are made at the same location across all available acquisitions. Analysis of optical imagery over the site confirms that the reported detection centroids are within ≈760 m of a radar site.

These examples demonstrate repeated detections by the EAR methodology and also highlight the potential to leverage characterize the "persistence" of specific RFI sources.

B. Source Polarization

NISAR's dual- and quad-polarization modes (DH, DV, and QP in Fig. 2) can also provide information on source polarization properties. For the example QP acquisition over Iowa, USA, depicted in Fig. 6, σ^0_{hh}, σ^0_{hv}, σ^0_{vh}, and σ^0_{vv} are all available with VV and HV detected as containing RFI contamination by the EAR method. The source's polarized nature is evident when comparing σ^0_{hh} and σ^0_{vv} [Fig. 6(a) and (b)] in which detections are made for σ^0_{vv}, while σ^0_{hh} [plot (a)] shows no obvious indications of RFI. This is further supported by comparisons of σ^0_{vh} and σ^0_{hv} [see Fig. 6(c) and (d), respectively], where detections are made for σ^0_{hv} but not σ^0_{vh}. The ratio $\sigma^0_{hv}/\sigma^0_{vh} > 10$ at the interference source's location (for which the scene backscatter should otherwise be identical) suggests that this source is primarily V-polarized.

For this initial investigation, the "global" studies in Section IV-D are conducted using only σ^0_{hh} and σ^0_{hv} polarized measurements. Source polarization properties in SH and SV

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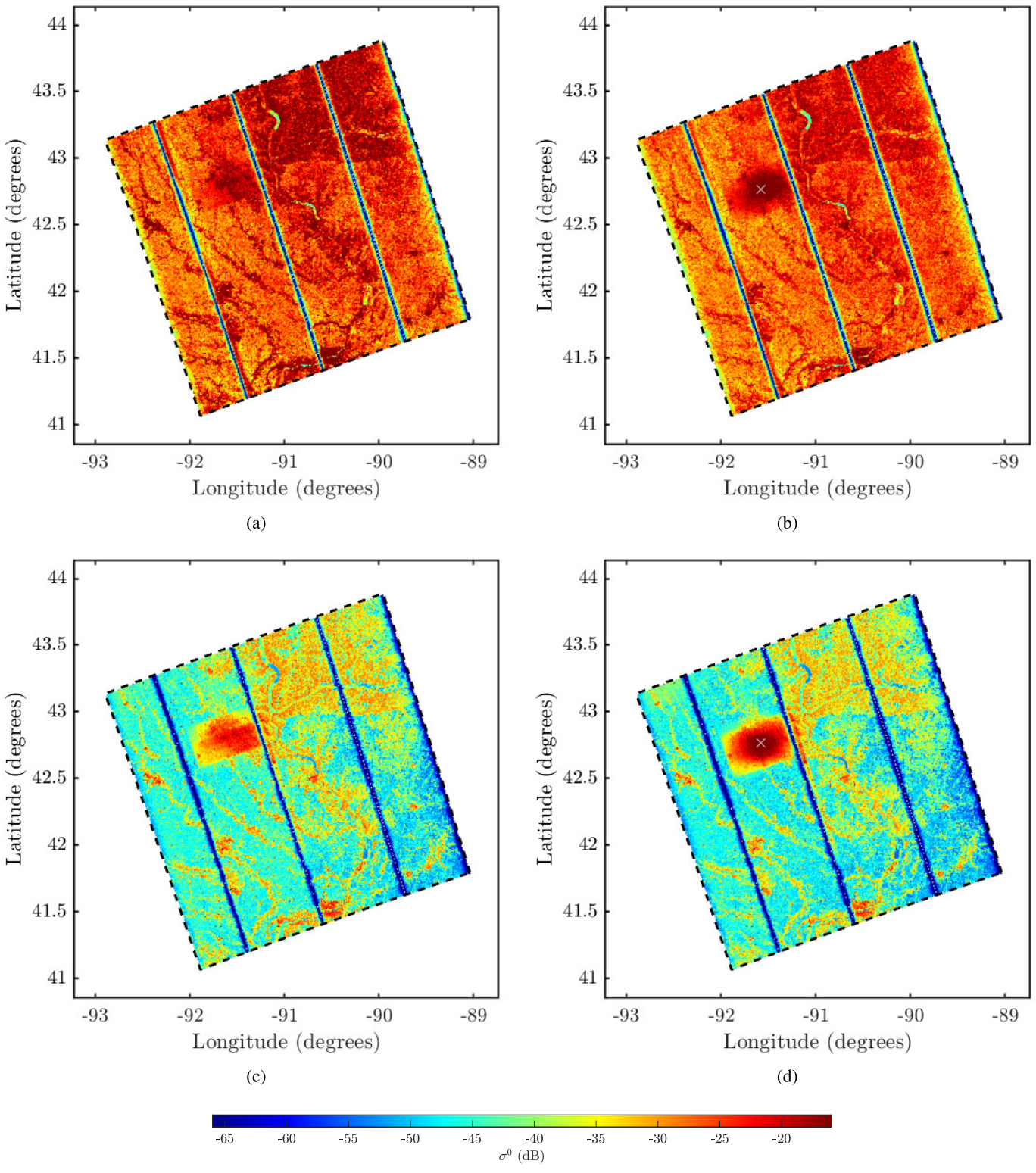


Fig. 6. Example QP acquisition over Iowa, USA, made on December 21, 2025 highlighting the capacity to couple EAR detection and geolocation outputs with NISAR's measurement mode properties to characterize the source's polarization properties. When the radar is operated in fixed PRF mode, gaps occur in the received range profiles when NISAR is transmitting. (a) σ^0_{hh}. (b) σ^0_{vv}. (c) σ^0_{vh}. (d) σ^0_{hv}.

mode data are uncertain given the availability of σ^0_{hh} and σ^0_{vv} measurements only. SH and SV acquisitions, however, represent only ≈6% of the February 2026 released data, so that information source polarization properties is frequently available.

C. Source Severity

Beyond detecting a source and identifying its location, it is also of interest to communicate intuitive and quantitative indicators of the extent to which RFI impacts a given

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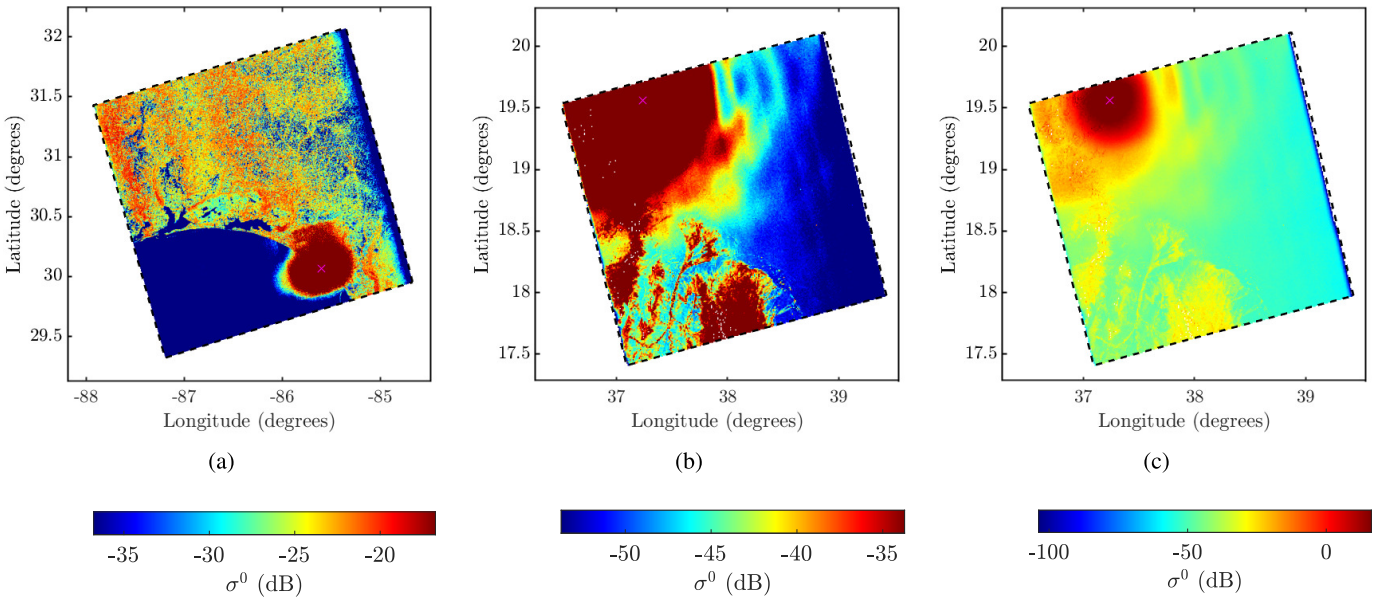


Fig. 7. Illustration of levels of corruption caused to acquisitions as indicated by the suggested severity indicator. (a) σ^0_{hv} for an acquisition over Panama City, Florida, USA, on December 21, 2025. Colorscale is limited to $\pm$10 dB relative to the scene's median. (b) σ^0_{hv} for an acquisition over Port Sudan, Sudan, on January 16, 2026. Colorscale is limited to $\pm$10 dB relative to the scene's median. (c) Same acquisition over Port Sudan with colorscale $\pm$60 dB relative to the scene's median.

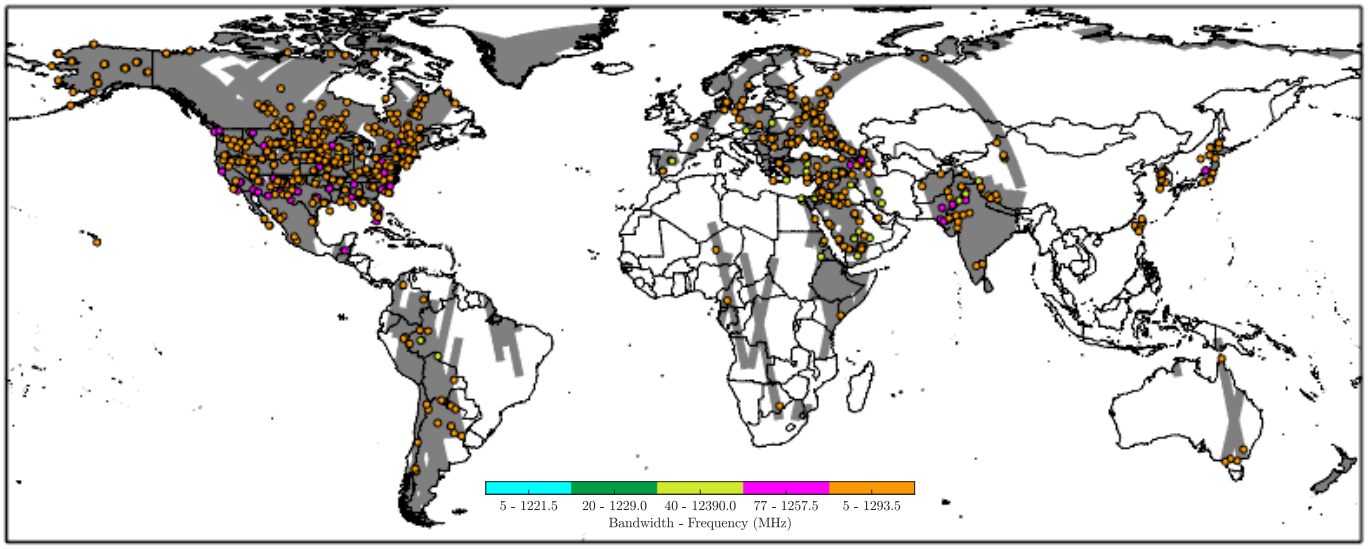


Fig. 8. Overall detections made across the 13 803 GCOVs binned by receiver bandwidth/frequency combinations.

acquisition. While measurement intensities at the locations of interference sources are expected to exhibit higher amplitudes, the extent to which they deviate from a scene's "background" level will depend on a wide range of factors. Those relate to measurement polarization, transmit power levels, transmit gain levels, RFI source properties, and the natural scene's scattering levels along with the range of potential calibration uncertainties impacting reported σ^0 levels. For example, across the EAR detections obtained in the February 2026 dataset, the reported σ^0 at the detection sites can range from as low as -40 dB to as high as 35 dB. Across the regions segmented but not detected by the EAR algorithm, the NRCS similarly ranges from -40 to 40 dB. The 98th percentile binarization prior to image segmentation ensures that the segmented locations have amplitudes greater than the scene's "background" σ^0 level, which itself may vary significantly from scene to scene. The "relative" nature of the EAR detector motivates the consideration of a more intuitive severity indicator ζ^s

$$\zeta^s = \frac{100\left(\overline{\sigma}^0_d - \sigma^0_{50}\right)}{\rho\sigma^0_{50}} \tag{4}$$

that compares the mean NRCS within the detection region $\overline{\sigma}^0_d$ to the scene's median σ^0_{50} normalized by a factor ρ to place ζ^s in the range 0%–100%. The factor ρ is polarization specific and is derived based on NRCS levels within segmented but RFI-free polygons. Use of the severity score ζ^s provides insight that is generalizable across scene scattering levels, polarization properties, and calibration uncertainties

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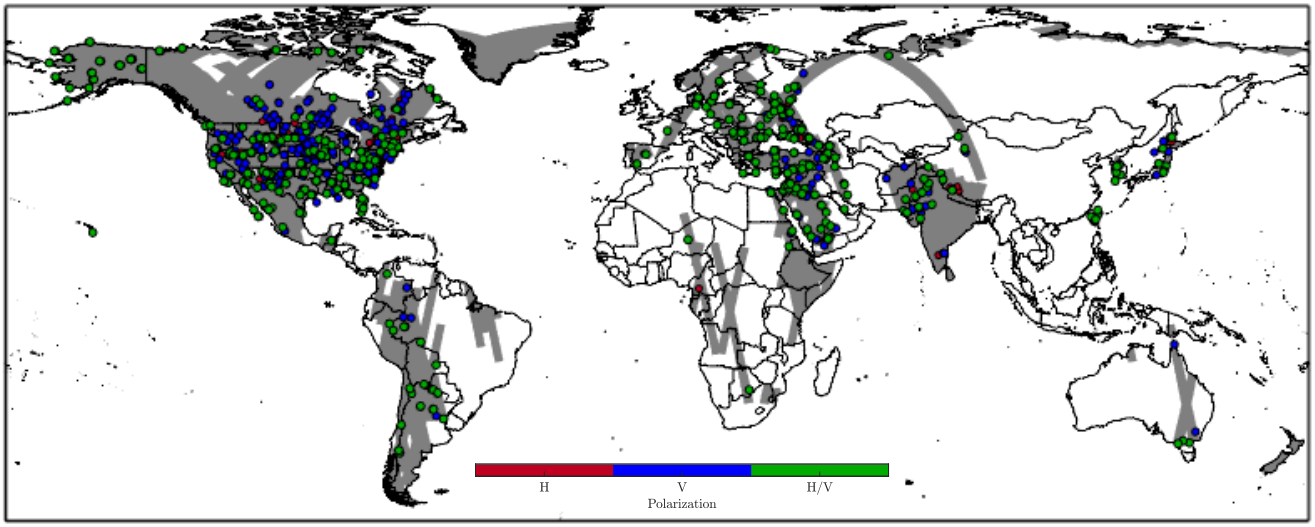


Fig. 9. Overall detections made across the 13 803 GCOVs binned by transmitter polarization signature. This excludes any SH and SV mode measurements.

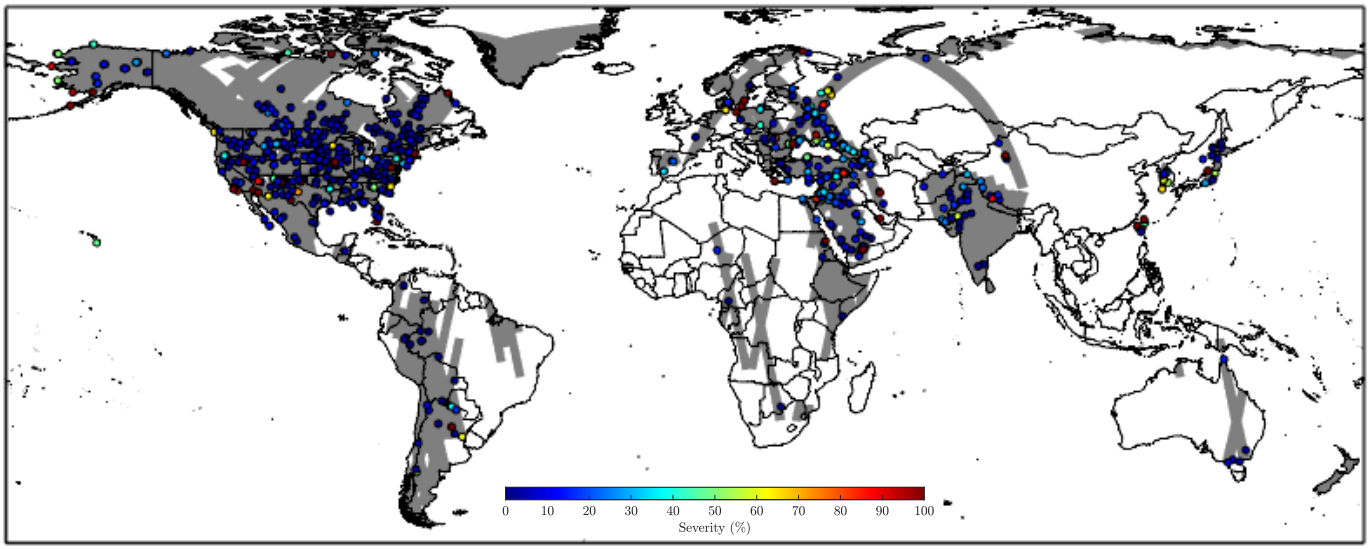


Fig. 10. Overall detections made across the 13 803 GCOVs binned by ζ^s.

since these factors should impact σ_d^0 and σ_{50}^0 similarly for RFI-free scenes. Any severity estimates that exceed 100% are saturated at the 100% level. This occurs for ≈10% of the measurements. In addition to allowing for reporting over a more intuitive 0%–100% scale, the score ζ is less dependent on the natural scattering levels of a given scene, polarization properties of a given acquisition or any calibration uncertainties as across most detections they would equally impact $\overline{\sigma}_d^0$ and σ_{50}^0. Examples of two acquisitions whose severity scores are $< 5\%$ and $> 90\%$ are depicted in Fig. 7(a) and (b), respectively. In both cases, the colorscale is limited to ±10 dBs relative to the scene's median. For the acquisition in Fig. 7(a) over Panama City, Florida, USA, the region impacted spans a 51 × 76 km extent. In contrast, for the acquisition over Port Sudan, Sudan, the region impacted by the interference source in Fig. 7(b) spans a 97 by 145 extent and it is only when the colorscale is adjusted to ±60 dB relative to the median in Fig. 7(c) that the contours of the interference source become discernible.

D. Global Trends

The EAR detection and geolocation methodology is applied to all 13 803 GCOVs where a total of 2396 detections are made. Their spatial distributions binned by $\mathcal{M}$ configuration, polarization, and severity are depicted in Figs. 8–10, respectively. A large number of detections are consistent with known locations of interference sources. An example includes the distant early warning (DEW) Line radar system along the Alaskan and Canadian Arctic coastlines. This also includes the majority of FAA long-range radar systems. It is, however, important to note that part of the "value added" by the proposed methodology is its identification of a range of sources whose properties are not known to be documented in the relevant literature, particularly across areas not in North America.

The trends in Fig. 8 suggest that 8.31%, 19.91%, and 71.78% of detections correspond to the $\mathcal{M}_2$, $\mathcal{M}_4$, and $\mathcal{M}_5$

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measurement configurations. While no detections are made in the $\mathcal{M}_1$ and $\mathcal{M}_4$ bands, it is important to interpret this finding in the context of the spatial trends discussed in Fig. 2. For the $\mathcal{M}_1$ and $\mathcal{M}_4$ bands, coverage is largely limited to uninhabited islands in the Nunavut Arctic Archipelago, the Russian coastline and an area of Greenland south of Pituffik Space Base (formerly Thule Air Base) where more potential interference sources may be expected. Additionally, the trends in Fig. 9 suggest that up to 73.46% of interference sources are dual-polarized, while 5.79% and 20.73% may be H and V polarized, respectively. This excludes any SH or SV acquisitions.

Severity levels summarized in Fig. 10 suggest that 50% exhibit ζ^s of 16.47% or lower, while a ≈10% subset exhibits $\zeta^s > 90\%$ with some of the FAA long-range radar systems identified in Fig. 1 exhibiting the highest severity levels. Put together, the trends suggest that when $\zeta^s < 75\%$ the mean σ^0 mosaics fail to capture the presence of interference sources.

As the total number of detections are made using some observations with multiple overpasses over the same locations, detections across multiple polarizations corresponding to the same acquisitions and potentially detections across multiple frequencies at the same location, it is estimated that the 2396 detections originate from more than 500 unique transmitters.

V. Conclusion

An initial demonstration of the EAR RFI detection and geolocation methodology was made across NISAR's February 2026 publicly released GCOV granules. While it is more common to apply RFI detection methodologies to measured in-phase and quadrature components of the received signals prior to any integration or SAR processing, the availability of amplitude information only as part of the GCOV product necessitates exploration of less standard means of detection. The proposed procedure therefore recognizes that as a result of NISAR's novel SweepSAR mode of operation, most terrestrial interference signatures will be shaped by the antenna footprint on the surface and therefore manifest as EARs within the acquisitions. As a result, through simple image processing techniques, a wide range of sources can be detected based on the level of their "circularity" relative to other naturally occurring surfaces. Detections are later geolocated through a centroid finding algorithm weighted by the intensity of measured σ^0 within the detected regions. While the proposed framework is shown to be instrumental for the detection and geolocation of sources, the resulting segmentation of the contours of RFI related artifacts can also serve as a computationally efficient means for the creation of "contamination masks" for use by prospective end users. Additionally, while the primary detection observable used was based on a "circularity" test incorporating the ratio of the detected area relative to its perimeter, numerous other tests are also available. Those include the ratios of major to minor axis, the roundness test utilizing the ratio of the area to the length of a region's major axis, the Hough transform along with related elliptical adaptations to name a few. Future studies are encouraged to consider and assess any benefits to using one indicator compared to another.

Beyond detection and geolocation of sources, this study also demonstrated the capacity for leveraging NISAR's elaborate measurement taking capabilities for further characterizing the sources' properties. While spectral contents can only be assessed in terms of falling within a given measurement's bandwidth relative to a given center frequency, less ambiguous determinations of sources' polarization properties and severity expressed in terms of a normalized σ^0 relative to the scene's median were also demonstrated.

Investigations relating to determinations of "source persistence" were also demonstrated using examples where multiple acquisitions over the same scene were available. In addition to providing the opportunity to characterize the transient versus persistent nature of detected interference sources, this also offered the opportunity to assess uncertainties in determinations of persistent sources' locations. Across the examples explored, it is assessed that estimates of the sources' locations are consistent to within a margin of ≈1.5 km such that applications requiring finer repeatability are encouraged to explore other centroid estimation algorithms beyond those proposed in this work.

Additionally, while detection thresholds were selected for the purposes of demonstration based on preliminary tests across the February 2026 public data release further refinement of the detection thresholds as more NISAR data becomes available is encouraged. For the detection thresholds with $\odot \geq 0.75$, preliminary assessments suggest that false alarm rates are on the order of 1%–3%, while for the less conservative detection thresholds with $\odot \geq 0.5$, the false alarm rates are estimated to be on the order of 10%. Applications more sensitive to false alarms may choose to enforce more conservative $\odot$ thresholds or complement $\odot$ with other detection metrics. This can be further complemented with terrain maps that can aid in flagging potential false alarms over regions with more complex topography. Additionally, the impacts of varying the 98th percentile binarization threshold can also be examined.

It is also instructive to note that while the locations of a number of interference sources are known and made public particularly in North America, the proposed methodology reveals the locations of several hundred sources of interference across the globe. The vast majority of their locations are not known or reported, which in turn highlights the utility of this and related methodologies capable of detecting and geolocating sources without a priori information. The disparity in the number of hotspots, limited to 18, suggested initially in Fig. 1 compared to the hundreds of sources identified using the proposed techniques is reiterated.

Finally, it is emphasized that the February 2026 release comprises data that has not been fully calibrated and validated. Among the range of enhancements that are expected to be applied to future NISAR data releases is the application of a mission RFI detection and mitigation algorithm. As this may reduce the prevalence of interference signatures across the dataset, the proposed methodology may aid in providing independent assessments of the efficacy with which mitigation is achieved, provide source locations for pre/post mitigation comparisons, and aid in the identification of any residual interference signatures.

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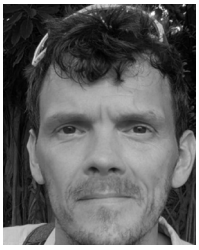
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